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Verfasser(in):	Kunz, Erik, Klasse F13Ia
Betreuende Lehrkraft	Frau Hecht
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Abbreviations

CIMP5: Coupled Model Intercomparison Project Phase 5

EDL: Electricité Du Liban

ICRC: International Committee of the Red Cross

IDF: Israel Defence Forces (*National Military of the State of Israel*)

KNMI: Koninklijk Nederlands Meteorologisch Instituut

LBP: Lebanese Pound

RCP: Representative Concentration Pathway

UN: United Nations

UNESCWA: United Nations Economic and Social Commission for Western Asia

UNICEF: United Nations Children's Fund

UNIFIL: United Nations Interim Force in Lebanon

WHO: World Health Organisation

Introduction

This paper investigates how and why water distribution in Lebanon has failed, and the implications of this failure for civilians, looking at a wide range of factors including both domestic mismanagement and international conflict, and details the outlook for the situation.

A portion of this paper discusses the current Lebanon-Israel and wider regional conflict, as this has had direct implications on water infrastructure in Lebanon. Given the political sensitivities of this topic, and the “fog of war” effect¹, this paper adopts a cautious approach to source selection, drawing on diverse sources including academic literature, international organization reports, independent Lebanese media, and documented field observations. Where sources present conflicting accounts, multiple perspectives are presented and critically evaluated.

This analysis is supplemented by field observations and interviews conducted in Beirut and southern Lebanon in September 2025, which aim to portray a more accurate picture of the situation on the ground.

¹ The ‘fog of war’ effect refers to the uncertainty, incomplete information, and confusion experienced during armed conflicts, which can obscure accurate reporting and decision-making.

1. Lebanon's Circumstances



Figure 1 - Political map of the Middle East and North Africa, highlighting Lebanon (Encyclopædia Britannica, Inc)

Water is a fundamental necessity for human life, and modern technological and institutional systems have made its provision nearly universal in most parts of the world. When scarcity occurs today, it is increasingly the result of human decisions and institutional breakdown rather than purely natural constraints.

Lebanon exemplifies this paradox. Despite having

comparatively abundant freshwater resources, the country faces chronic shortages and unreliable distribution. Years of economic collapse, compounded by recurring conflict, have crippled Lebanon's public water and wastewater systems (UNICEF Lebanon, 2025). Many households now depend on costly and often poor-quality private water trucking.

The problem does not lie in rainfall patterns as Lebanon receives some of the highest precipitation levels in the Middle East, to the point that it had gained the reputation as the “water tower of the Middle East” (Action Against Hunger et al., 2025). Rather, it lies in insufficient storage capacity, inadequate infrastructure maintenance, financial mismanagement, conflict, and fragile governance structures (World Bank, 2025). Public sector providers, unable to cover fuel and repair costs amid the national power crisis, frequently deliver less than the minimum 35 litres per person per day recommended for health and sanitation, forcing citizens to rely on inconsistent and expensive private alternatives (“UNICEF report,” 2022). Lebanon's case demonstrates how man-made dysfunction, rather than environmental scarcity, can render even water-rich societies unable to guarantee basic access for their populations.



Figure 2 - Thawra Protests in Lebanon (2019, Amir Makar/AFP via Getty Images/Newlines)

Lebanon's governments in modern history are marked by chronic weakness, fragmentation, and corruption. Political authority is divided among sectarian parties that often act more as patrons of their communities than as representatives of the state. This power-sharing system, formalized after the Lebanese

civil war, which ended in 1990, was meant to guarantee stability, but in practice it has created deadlock and clientelism that prevent coordinated policy (World Bank, 2022). As a result, the state struggles to provide reliable services such as electricity, waste management, and especially water distribution. These issues have led the country's population to hold extreme distrust towards the government and its institutions, which culminated in an attempted revolution from 2019 to 2021 (known as the "Thawra") (Nagle, 2024). This distrust in the state has further led people to find their own solutions and stop relying on its services. International institutions, including the World Bank, describe Lebanon as being in one of the most severe economic and governance crises of modern times. (World Bank, 2024a)

The origins of this paralysis lie in the aftermath of the 1975–1990 civil war, which destroyed much of Lebanon's infrastructure and left a political system built around sectarian quotas rather than functional institutions. The state has since remained fragmented, with armed non-state actors operating independently of government control. This structural weakness has been compounded by repeated Israeli military campaigns. In 2006, Israeli bombardments deliberately targeted bridges, fuel depots, power plants, and water facilities, crippling basic services for months. ("Hezbollah Conflict in 2006 | How does law protect in war? - Online casebook," 2006) Since 2023, similar strikes have destroyed residential areas, hospitals, and critical infrastructure in southern Lebanon and Beirut's suburbs ("Lebanon," 2025) These repeated campaigns

have systematically undermined the state's already limited ability to guarantee water distribution.



Figure 3 - Electricité du Liban building, heavily damaged by the 2020 port blast (2025, Erik Kunz)

The economic collapse that began in 2019 has amplified the crisis. The state defaulted on its debt in 2020, the currency lost over 90 percent of its value, and inflation eroded the operating budgets of public utilities. (World Bank, 2024a) The 2020 Beirut port explosion further destroyed water, electricity and other infrastructure across the capital and displaced hundreds of thousands. (Majzoub, 2021) Figure 3 shows an example of this, the management building of EDL, Lebanon's national electricity provider, which was badly damaged in the blast and as of

September 2025 has been left in disrepair, not indifferent to the state of the agency itself, which fails to provide a reliable provision of electricity for the country (Klinken, 2021). Today, this combination of internal mismanagement, external attacks, and fiscal collapse leaves water systems vulnerable to contamination, intermittent supply, and dependence on external power generators. What emerges is not a single cause but an accumulation of failures, with conflict, domestic corruption, and economic breakdown reinforcing one another to leave Lebanon's population exposed to recurring failures in water provision.

2. Mechanisms of Failure in Water Distribution

2.1. Institutional Weakness and Infrastructure Degradation

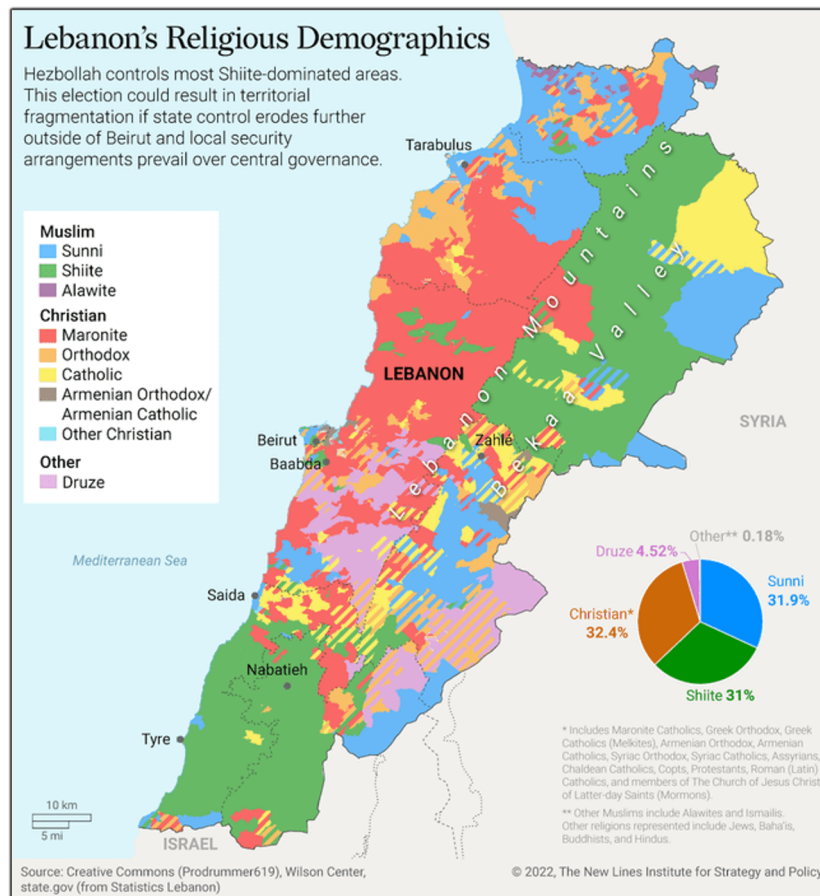


Figure 4 - Map depicting the religious demographics of Lebanon (2022, The New Lines Institute for Strategy and Policy)

Lebanon's political system is defined by its sectarian power-sharing structure, designed to accommodate a multi-faith society. The country's three largest religious groups — Shi'a Muslims, Sunni Muslims, and various Christian denominations — each represent roughly a third of the population, with smaller minorities like Druze and Alawites making up the remainder.

Figure 4 displays how diverse and intertwined

the population is religiously. Political affiliation is often determined less by policy but by sectarian identity, with many citizens voting along family or village lines. For example, in some towns residents reported the mayor of a village having had the role for decades, leading to them having become part of local families, having watched generations grow up and helped families through crises and wars. Residents indicated they continue to vote for this person for personal reasons, and not because of the ambitions of the political party he may belong to.²

² Comments of local citizens in Yanouh and Bazouriye, South Lebanon, in September 2025.

This entrenched loyalty reinforces fragmentation not only between sects but also within them, perpetuating political rigidity and deadlock. ([“Lebanon’s politics and politicians | Chatham House – International Affairs Think Tank,” 2023](#)). With the goal of preventing another civil war, it was ensured that no sectarian group could hold absolute power over the government. This however means that, when no compromise can be found, institutional reforms stall indefinitely.

Beyond that, Lebanon faces a severe energy crisis. The country’s national energy provider, Electricité du Liban, has since 2020 been unable to supply the country with adequate electricity. The limited supply is divided by region in the form of allocated time slots, whereby one town might receive grid power for a couple hours, and then the next, and so on, though even these allocated slots prove unreliable. Even critical services, like hospitals, government institutions, and the country’s only commercial airport in Beirut, are not supplied a reliable flow of power. This has led most private households to rely on storage batteries and solar power, while those with larger power requirements such as hotels, large apartment blocks, offices and commercial sites rely on expensive diesel power generators ([Hallak et al., 2024](#)).

The result of this is drastic, with noise and air pollution from generators overpowering the atmosphere in the capital of Beirut ([Fakhri et al., 2025](#)) ([Hallak et al., 2024](#)), and sudden power outages having become standard, but one of the biggest impacts is that on water filtration facilities and pumps. These, generally already underfunded and unable to afford a third-party power source, are often left standing still due to the lack of grid power available ([“Lebanon’s water supply system,” 2021](#)).

This systemic paralysis has had profound consequences for the management of essential services, including water distribution. Public institutions responsible for water are fragmented between regional water establishments and overlapping ministries, yet none hold the authority or capacity to implement long-term reforms. As a result, investment in infrastructure is chronically delayed, maintenance is neglected, and emergency solutions such as private water trucking have become the default system for large parts of the population. Without the political will or institutional stability to

coordinate strategy across these sectarian lines, Lebanon's water sector reflects the broader dysfunction of the state itself. It is fractured, underfunded, and dependent on privatized solutions rather than a coherent national framework.

This mismanagement, paired with leaks caused by the inadequate maintenance of existing water infrastructure, lead the government-run central water management firm to lose around half of its water supply (Yared, 2023), further straining an already crippled water distribution situation.

2.2. Financial Collapse

In part due to the prior political collapse, Lebanon's economy came crashing down with it. Having lost roughly 98 percent of its value since October 2019 (Homsy, 2023), the Lebanese Lira has suffered some of the worst devaluation of any modern currency. This intense hyperinflation over such a short time period has had a detrimental effect on all aspects of Lebanese life, and has also meant a significant dip in funding and therefore the capabilities of local authorities, as they, unlike some private sectors, were not capable of diversifying their income to other currencies like the Euro or US Dollar, meaning they were stuck functioning under the almost valueless Lebanese Lira.

Compounding the problem, the official exchange rate maintained by the Banque du Liban remained artificially close to its pre-2019 level, drastically underrepresenting the Lira's real depreciation. Most private business and transactions are conducted either in US dollars or in Lebanese lira valued at the parallel (informal) market exchange rate rather than the official rate. Accordingly, trade is commonly carried out using a mix of USD and lira converted at this rate.³ As a result, Lebanon effectively operates with dual exchange rates: a parallel, market-determined rate used by the private sector, and an official rate applied to public budgets. This discrepancy severely diminishes the real value of municipal budgets, as allocations denominated in LBP no longer correspond to actual purchasing power in goods, services, or infrastructure. (Hausmann et al., 2023)

³ Field observations in Lebanon, September 2025

The chronic underfunding of public services that ensued has exacerbated the issue caused by the political instability and corruption whereby there is a significant lack of new development projects and funding for the maintenance of existing infrastructure.

2.3. Conflict

Lebanon has endured repeated wars that have made destruction a recurring feature of its modern history. This persistent instability has severely constrained long-term development and investment in infrastructure.

The civil war (1975-1990) left the country's water systems "barely operational" (Gharios et al., 2021). While the civil war ended in 1990, southern Lebanon in particular has remained a zone of conflict since. In 1985, Hezbollah emerged out of small militia groups in response to Israel's repeated invasions and occupations of the south ("The Hizballah Program - An Open Letter," 2006). This political party and armed group succeeded in forcing the IDF out of southern Lebanon but sparked a new phase of conflict as the IDF have since conducted repeated attempts at re-gaining their once held control of the area.

This phase of the conflict culminated in 2006, when, over the span of four weeks, Israeli forces conducted a large-scale assault across Lebanon using both ground and arial bombardments that catastrophically destroyed the country's infrastructure. Israeli forces are reported to have "pounded buildings into the ground, reducing entire neighbourhoods to rubble". (Amnesty International, 2006).



Figure 5 - Memorial of Hezbollah soldiers killed in the current conflict with Israel in Dahiyeh, Southern Beirut, Lebanon (2025, Erik Kunz)

Since October 2023, the conflict in the Gaza strip has caused a major spillover of instability and military conflict across the region, including in Lebanon, where Hezbollah's armed wing is engaged in frequent armed conflict with the IDF. 2024

saw the start of the sixth Israeli invasion of Lebanon (Gharios et al., 2021), driving Israel's military actions in Lebanon to new heights, with renewed airstrikes across the country as well as an increasing occupation of southern Lebanese towns by IDF forces. While a ceasefire has been in place since November 2024, UNIFIL claim Israel to have violated it over 10000 times as of November 2025 ("Year after ceasefire, peace eludes south Lebanon as Israeli strikes continue," 2025). This new conflict has left much of southern Lebanon unstable and partially destroyed. Figure 6 displays water pumping facilities in south Lebanon that have suffered at least moderate damage since October 2023 due to the conflict.

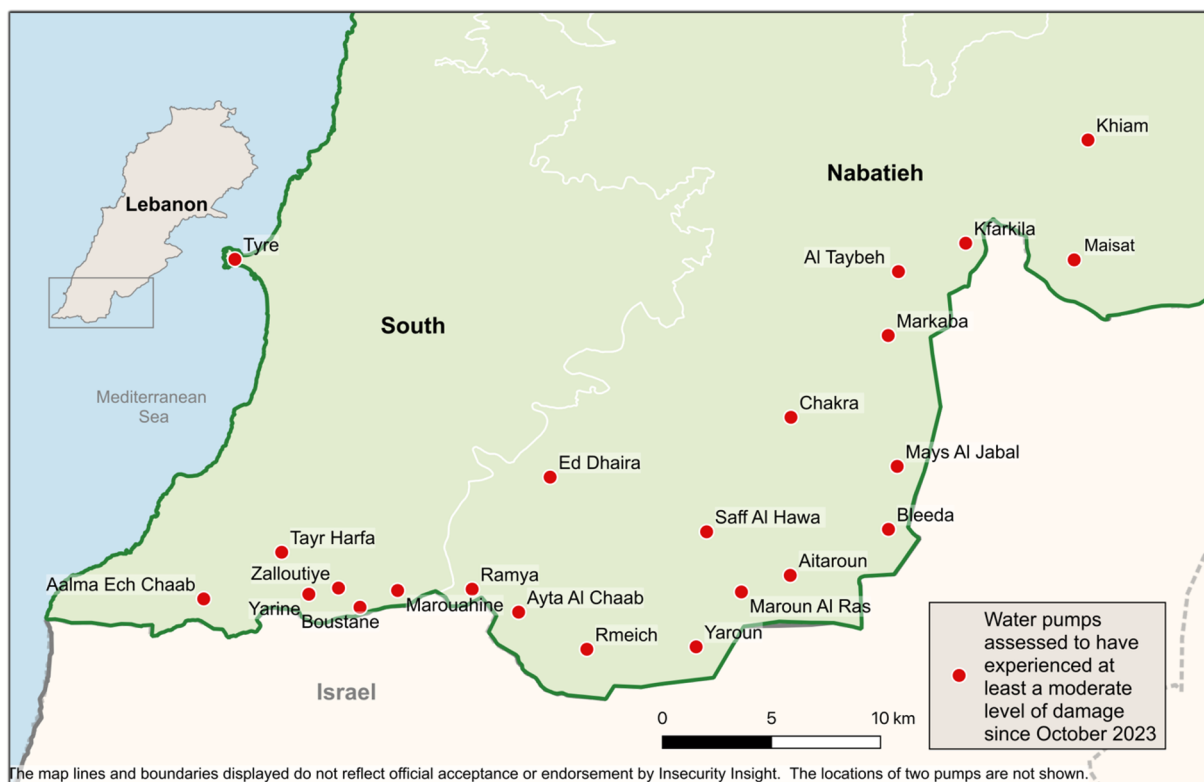


Figure 6 - Water pumping facilities in South and Nabatieh governorates assessed to have sustained at least moderate damage since October 2023 (2025, Action Against Hunger et al.)

These water pumps were responsible for the distribution of running water amongst many of the surrounding settlements, and the damage inflicted upon them has caused frequent widespread disruption to water distribution ("Not one drop of water' in nine south-east Lebanon villages after Israeli strike disrupts water infrastructure," 2024). A report by UNICEF states that, as of October 2024, at least 28 water facilities had been damaged by the recent conflict, affecting water supply to over 360,000 people, and

highlights the larger, hidden damage caused by the inaccessibility of affected areas to technical teams (“Increasing bombardment damaging essential services and putting children in Lebanon at great risk,” 2024).



Figure 7 - Bomb damage to a residential building in Tyre, South Lebanon (2025, Erik Kunz)



Figure 8 - Bomb damage to a water pumping facility near Tyre, South Lebanon (2025, ICRC)

The severe impact the conflict has on water distribution is not only due to the destruction of water pumping facilities themselves but also includes frequent Israeli bombardment of a wide range of facilities necessary for powering water infrastructure and rebuilding destroyed infrastructure. Israeli airstrikes have repeatedly hit such key facilities, directly preventing Lebanon from rebuilding and solving the water crisis. One such example is a wave of airstrikes on the 16th October 2025 which destroyed the South Lebanon Water Establishment’s fuel storage facility, which contained half a million litres of diesel used to power water stations and wells across the south of the country.

القصف الإسرائيلي يدمر مخزن المحروقات الاستراتيجي لمؤسسة مياه الجنوب) [Israeli bombing destroys the strategic fuel depot of the South Water Corporation],” 2025). The same wave of airstrikes destroyed one of Lebanon’s largest industrial sites, a cement and asphalt factory, further limiting the nation’s reconstruction capabilities. Israel claims the strikes to have only targeted Hezbollah infrastructure. (“Series of Israeli strikes across south Lebanon, Israel claims Hezbollah targets,” 2025)

Air and drone strikes in the village of Yanouh, for example, which is home to a large natural water source that prior to 2023 supplied much of the surrounding areas with running water, destroyed the town’s pipes and infrastructure, resulting in a village



Figure 9 – Noted locations in South Lebanon on a map (2025, Google Maps, Erik Kunz)

whose inhabitants are left to rely on large water bottles for everything from showering to cooking.⁴ In nearby Qana, much of the city centre had been destroyed leaving many of its citizens internally displaced, and resulting in similar situation of water unavailability in the town.⁵ This situation is not unique, but has rather become the

status-quo for much of Lebanon’s south since the beginning of this new phase of war in late 2023. In total, within 5km of the border with Israel, an estimated 90% of water services have been disrupted by the conflict (“Six Months After Ceasefire, Hundreds of Thousands in Lebanon Still Deprived of Running Water - Action contre la Faim,” 2025).

2.4. Geographical and Environmental Pressures

Lebanon’s water distribution system is severely strained due to decades of inadequate environmental regulation, weak enforcement, and the absence of effective waste management infrastructure. This has resulted in untreated waste accumulating in natural environments, contaminating rivers, lakes, and coastal waters, and rendering many water sources unsafe or unusable (Arab NGO Network for Development, 2025). The interaction between long-term structural neglect and accelerating climate pressures has produced severe environmental degradation, significantly reducing the country’s effective natural water availability.

Waste management represents a particularly acute challenge. In this context, widespread littering observed across much of Lebanon should not be interpreted as a cultural practice, but rather as a behavioural consequence of prolonged crisis conditions.

⁴ Field observations and interviews in Yanouh and Tyre, South Lebanon, September 2025

⁵ Interview with resident in Qana, South Lebanon, September 2025

Conversations with residents consistently revealed a sense of exhaustion and resignation rooted in decades of war, economic collapse, and political instability⁶. For many, daily life is dominated by immediate survival rather than long-term environmental responsibility. This collective fatigue has contributed to a visible erosion of public care for shared spaces, including widespread waste disposal along roadsides and pavements, further intensifying the pollution of natural water sources.

Empirical studies confirm that this degradation is not merely perceptual but measurable. A WHO-published water quality assessment found that over 90% of collected wastewater in Lebanon is discharged untreated into rivers, with 60% of river samples exceeding safe levels for *E. coli* and faecal coliforms. This contamination directly compromises the suitability of these waters for irrigation, livestock use, and human contact (El Chamieh et al., 2024). Similarly, a nationwide microbial survey across 14 rivers detected faecal bacteria in over 95% of samples, with the majority exceeding safety thresholds for irrigation and recreational use, underscoring the systemic degradation of Lebanon's freshwater systems (Dagher et al., 2021).

These structural failures are further exacerbated by climate change. Middle Eastern countries are among the most vulnerable globally to climatic shifts (Christensen et al., 2013), and Lebanon has experienced a marked decline in precipitation. Rainfall levels have fallen by approximately 50% compared to the annual average ("Lebanon's Agricultural Catastrophe," 2025). The Litani River, the country's longest and most critical freshwater resource, recorded an almost 50% reduction in water levels in 2025 compared to 2024 (Mansour, 2025). Concurrently, declining winter snowfall has increased forest exposure to wildfires and further weakened environmental resilience (Mansour, 2025).

The cumulative impact of these pressures became especially visible during the summer of 2025, when Lebanon experienced a severe drought. Water levels in major natural reservoirs, including Lake Qaroun in eastern Lebanon, fell to critically low levels, as

⁶ Comments of residents in Beirut, Lebanon in September 2025



Figure 10 - Lake Qaraoun during the Drought in 2025 (2025, Mercy Corps Lebanon)



Figure 11 - Lake Qaraoun at normal water levels in 2024 (2024, lebanonuntravelled.com)

illustrated in the accompanying images. Hydropower plants were forced to suspend operations due to insufficient water volumes ([“Lebanon’s worst drought on record drains largest reservoir,” 2025](#)), exacerbating the country’s existing electricity crisis. At the same time, Lebanon’s agricultural sector suffered substantial losses as inadequate water storage and distribution infrastructure proved incapable of buffering climatic shocks.

Overall, Lebanon’s geographical and environmental pressures on water availability arise from a compounding interaction between environmental contamination, weak regulatory enforcement, and climate-driven hydrological decline. Untreated wastewater and widespread pollution have rendered large portions of surface water unusable at the point of extraction, while declining rainfall, reduced snowpack, and falling river levels have progressively diminished the volume and reliability of available freshwater inputs. These pressures function as upstream constraints on the water distribution system, degrading source quality and reducing supply capacity before issues of allocation, infrastructure, or consumption are even encountered.

3. Consequences

3.1. Economic Impact and Social Inequality

Figure 12 - Affluent "Saray District" in central Beirut, boasting modern architecture, luxury brand outlets and updated infrastructure (2025, Erik Kunz)



Figure 13 - Refugee Camp in Beqaa Valley, near the Syrian border, Eastern Lebanon, one of many camps home to hundreds of thousands of Syrian refugees. (2025, Erik Kunz)

In the absence of a regular supply from formal water pipes, civilians resort to costlier alternatives such as private water trucking and bottled water. The impact is stark. Surveyed residents in Beirut reported an average yearly water cost of 2124 USD per year in 2022 ([Choueiri et al., 2022](#)), more than the median Lebanese household income of just 1464 USD per year in 2023. ([Human Rights Watch, 2023](#))

Lebanon is one of the most unequal countries in the world in terms of income distribution (UNESCWA, 2022). Further worsening the situation, this severe inequality is emphasised by a stark lack of upward mobility (UNESCWA, 2022). Lebanon is also home to an incredibly large number of refugees, mostly from neighbouring Palestine and Syria. These refugees are not granted the same rights as Lebanese citizens and therefore suffer from even more unequal circumstances than the worse-off of the Lebanese (Arab NGO Network for Development, 2025). Refugees from Syria are mostly employed in low-skilled jobs while Lebanese are more frequently in skilled positions (World Bank, 2024b). The lack of an official census in Lebanon since 1932 leads to a lack of precise figures, but estimates place the number of refugees of Palestinian origin at anywhere from 250,000 to 500,000 (“Palestinian refugees in Lebanon,” n.d.), and of Syrian at an estimated 1.5 Million (UNHCR Lebanon, 2024). Taking into consideration the country’s population of Lebanese citizens of only 4 Million (“moph,” 2024), its small size, and already crippled state, taking care of the needs of such a huge proportion of refugees is a near impossible task. The result is an impoverished and unsupported refugee population living without critical necessities.

Paired with approximately 900.000 internally displaced people (“Lebanon | OCHA,” 2025), mostly due to the conflict with Israel in the south, Lebanon’s civilians lack any of the structure and social support needed to withstand the many crises the country faces. The failing state struggles to keep itself standing, and with millions of displaced and impoverished people, whether Lebanese or not, it is unable to support them adequately, making the impact of the water crisis on this part of the population disproportionate.

3.2. Agriculture and Food Security

The combined effects of water scarcity, excessive agrochemical use, and declining soil fertility contributed to a significant contraction of agricultural land. Satellite imagery analysis indicates a nationwide reduction in cropland cover of 52% in 2025 compared to 2019 (Lebanon Crisis Analytics Team and Mercy Corps, 2025) highlighting the extent to which environmental and institutional pressures now structurally constrain Lebanon’s water and food security.

Reduced agricultural output has, in turn, increased Lebanon's dependence on food imports and exposed the population to escalating food prices. Rural communities, whose livelihoods are closely tied to farming, have been particularly affected, as unstable water access has increased production risks and reduced income reliability. This erosion of agricultural resilience has weakened local coping mechanisms and amplified existing economic and social vulnerabilities, especially in already marginalised regions (ESCWA, 2020). Over 82% of farmers interviewed in South Lebanon reported being unable to get enough water to irrigate crops and supply livestock. ("Six Months After Ceasefire, Hundreds of Thousands in Lebanon Still Deprived of Running Water - Action contre la Faim," 2025) Many farmers have also left their land due to the dangers of the conflict and the destruction of infrastructure, with an estimated 92,000 people who used to live in Lebanon's southern border regions having left and not returned since October 2023 ("Six Months After Ceasefire, Hundreds of Thousands in Lebanon Still Deprived of Running Water - Action contre la Faim," 2025).

3.3. Health and Sanitation

The lack of reliable water carries severe health risks, especially for those unable to routinely afford private or trusted bottled water supplies. Vulnerable communities are forced to resort to unsafe, in cases contaminated water. ("Six Months After Ceasefire, Hundreds of Thousands in Lebanon Still Deprived of Running Water - Action contre la Faim," 2025). Health facilities themselves are often affected by water shortages, undermining basic sanitation standards and limiting their ability to provide effective care. This compounds public health risks during periods of crisis, when medical systems are already overstretched and under-resourced.

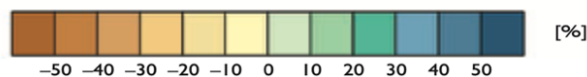
Over time, persistent deficiencies in water and sanitation systems contribute to broader structural health vulnerabilities. Chronic exposure to unsafe water disproportionately affects children, the elderly, and displaced populations, leading to long-term developmental, nutritional, and immunological consequences. As a result, water insecurity in Lebanon functions not only as an immediate humanitarian concern but

also as a systemic public health challenge that exacerbates existing socio-economic inequalities.

4. Outlook

4.1. Climate Change

mean rcp85 relative precipitation 2081–2100 minus 1986–2005 Jan–Dec AR5 CMIP5 subset



mean rcp85 temperature 2081–2100 minus 1986–2005 Jan–Dec AR5 CMIP5 subset

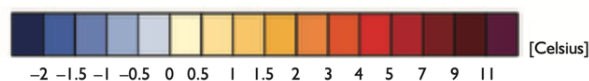
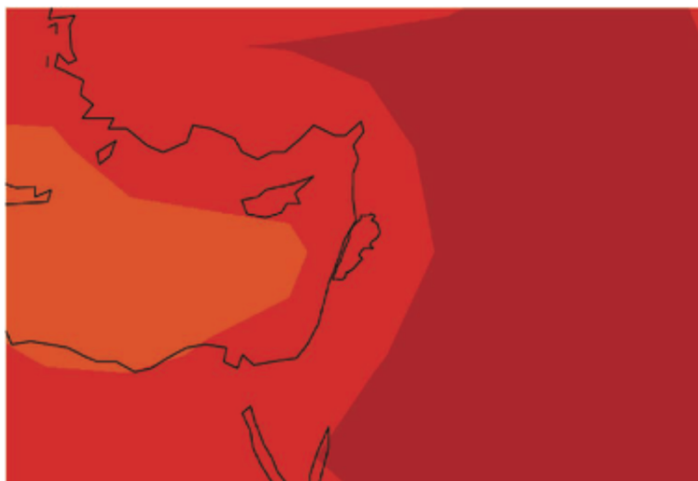


Figure 14 - CMIP5 Projections of Mean Annual Relative Precipitation Change for 2081–2100 Compared with 1986–2005, RCP8.5 Scenario (KNMI Climate Explorer)

Figure 15 - CMIP5 Projections of Mean Annual Temperature Change for 2081–2100 Compared with 1986–2005, RCP8.5 Scenario (KNMI Climate Explorer)

Climate models can be used to determine future scenarios in which Lebanon may find itself. The examples in Figures 14 and 15 show that, according to the Representative Concentration Pathway 8.5 Global Warming Scenario, which estimates what would happen without efforts to prevent climate change, (Harrisson, 2019) Lebanon will face a

significant drop in precipitation of minus 20 to 30% and an increase in mean temperature of between 4 and 5 Celsius by the end of the 21st century. This becoming reality would prove challenging for Lebanon and requires significantly stronger infrastructure to reliably withstand.

4.2. Moving Forward

This long-term hurdle further emphasises the urgency of solving Lebanon's water crisis before it worsens. Providing water security requires managing a multitude of crises and cannot be solved by Lebanon alone. Building on the analysis presented in Chapters 1 to 3, there are three main issues that need to be solved to be able to move forward. These are:

- i. Conflict with Israel

Lebanon's persistent state of conflict with Israel has significantly constrained the country's long-term development, including its ability to address critical infrastructure challenges such as water security. Recurrent military tensions, the legacy of past conflicts, and the ongoing risk of escalation create an environment of instability that discourages investment, limits regional cooperation, and diverts political and financial attention away from civilian infrastructure.

This conflict cannot be viewed in isolation. Lebanon is deeply embedded in wider regional dynamics, particularly the unresolved Israeli-Palestinian conflict and broader power struggles involving Iran and its regional allies, including Hezbollah. As long as Lebanon remains entangled in these conflicts, prospects for sustained economic recovery, international support, and cross-border resource cooperation remain limited. Consequently, addressing Lebanon's water crisis is closely linked to broader regional stability, which cannot be achieved through military means alone.

ii. Domestic Politics

Internally, Lebanon's political system presents a major obstacle to effective water governance. Chronic corruption, weak institutions, and political fragmentation have undermined the state's capacity to plan and implement long-term infrastructure projects. Decision-making is often driven by short-term political interests rather than national priorities, resulting in fragmented policies and inconsistent enforcement.

Additionally, sectarian and identity-based politics further complicate reform efforts. Public resources are frequently distributed along political or communal lines, reducing efficiency and accountability. In the context of water management, this has contributed to the uneven service provision, neglected maintenance, and reliance on informal or private alternatives documented in this paper. Without substantial political reform and stronger institutional coordination, technical solutions alone are unlikely to produce lasting improvements.

iii. Financial Resources

Finally, the issue of financial resources represents a decisive constraint on any long-term solution to Lebanon's water crisis. Even where technical expertise and potential policy frameworks exist, Lebanon lacks the fiscal capacity to invest in, maintain, and modernise critical water infrastructure. Years of economic collapse, public debt, and currency devaluation have severely limited state spending, while public trust in financial institutions has eroded.

This financial weakness does not exist in isolation. Ongoing regional conflict discourages foreign investment and long-term international funding, while domestic political fragmentation undermines transparent allocation and effective oversight of available resources. As a result, funding that is secured is often short-term, fragmented, or poorly implemented, preventing the development of resilient, nationwide water systems.

Without financial stability and credible governance structures, efforts to address water scarcity risk remaining reactive rather than strategic. Climate change will further intensify these pressures, increasing costs while reducing the margin for error. Addressing Lebanon's water crisis therefore depends not only on technical solutions, but on restoring the economic and institutional capacity required to sustain them.

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Hiermit erkläre ich an Eides statt, dass ich die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe. Alle Stellen meiner Arbeit, die im Wortlaut oder dem Sinn nach anderen Werken entnommen sind, habe ich in jedem Fall unter Angabe der Quelle als Zitat bzw. als Entlehnung kenntlich gemacht. Die vorliegende Arbeit hat in dieser oder einer ähnlichen Form noch keiner anderen Prüfungsbehörde vorgelegen.

Nürnberg, 13.01.2026

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